



Sharks live much longer

Major review of studies on shark lifespan indicates that they can live for twice as long as previously estimated. <http://digit.in/OldSharks>

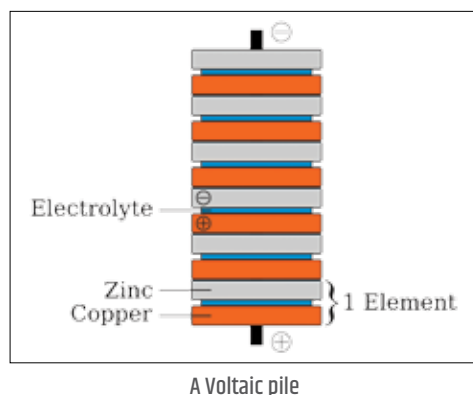


Face ID is just the beginning

Future phones may go all out with biometrics such as facial, iris and fingerprint recognition. <http://digit.in/MoreBio>

ried out an experiment in 1791 that led to the invention of the Voltaic cell.

The Voltaic Pile, credited as the first true battery, consisted of layers of copper and zinc divided by layers of cardboard or cloth soaked in brine. While it was the first battery that could provide consistent current, a few drawbacks like electrolyte leakage and hydrogen



bubble formation made it unfeasible as a power source. In 1802, William Cruickshank designed the first electric battery for mass production. He arranged square sheets of copper with equal-sized sheets of zinc and placed them into a long rectangular wooden box and soldered them together. While grooves in the box held the metal plates in position, the box itself was filled with an electrolyte of brine, or watered-down acid. This resembled the flooded battery that we still use today and solved the leakage issues with the Voltaic pile to great extent.

Interesting fact – One of the words for a battery in French is 'pile' (pronounced peel)

While the Daniel Cell addressed the hydrogen bubble problem with the Voltaic pile, it was still a primary cell – a type of battery that is of no use once the chemical reactions within it are completed and the metal elements are exhausted. Gaston Planté addressed this problem with the invention of the lead acid battery – a type of apparatus which could be 'charged' by passing an electric current opposite to the one it produces. Lead acid batteries are still used in automobiles.

Interesting fact – One volt is quantified as the voltage produced by one Daniel cell

PORTABILITY

Anyone working on battery technology primarily has simple goals – how to make smaller, lighter batteries that can provide more energy than what has been achieved so far. The goals weren't much different back in 1866, when Georges Leclanché developed a battery with a zinc anode and a manganese dioxide cathode wrapped inside a porous material, with ammonium chloride as the electrolyte. While this design itself showed longer shelf life, what made it significant was a version developed by Carl Gassner in 1886 which used a paste-like material for the electrolyte, making the design greatly portable. The original ammonium chloride was mixed with Plaster of Paris to get this paste. While this model was improved by

alterations to the electrolyte type and structure, as well as other elements in the mix, the Gassner model to this day is considered responsible for modern day portable batteries and is almost directly credited for the invention of the flashlight.

LITHIUM THE SAVIOR

At the turn of the 20th century, electricity was the talk of town. Don't throw us into the loony bin if we tell you that electric cars were more common in the early 1900s than they are today – because they were. Around that time, about 40 percent of American automobiles were powered by steam, 38 percent by electricity, and 22 percent by gasoline. 33,842 electric cars were registered in the United States between 1900 and 1910. It was the unreliability of the batteries in use, along with a couple of factors like limited range, that brought about its demise as gasoline, and gas powered vehicles grew in popularity.

After almost a century of incremental improvements through Nickel-Cadmium, battery technology was still at a troublesome junction – with Nickel's environmental effects also creating a growing need for a better alternative.

Enter Lithium, the metal with the lowest density and highest energy-to-weight ratio. While the first Lithium batteries came into the market as early as 1970s, it was three important developments in the 1980s that brought about the currently popular lithium ion batteries, involving the discovery of better electrodes. This led a research team managed by Akira Yoshino of Asahi Chemical, Japan, to build the first lithium-ion battery prototype in 1985, a rechargeable and more stable version of the lithium battery. Sony commercialized the lithium-ion battery in 1991.

THE BATTERY OF THE FUTURE

The race around the world to discover the battery of the next generation is charging ahead. While Li-ion batteries are doing just fine as the portable batteries of choice, recent spates of battery explosions – most notably with the Samsung Galaxy Note 7 – have damaged our trust in Li-ion.



The 1902 Baker Electric, a popular electric car in its time

Everything from gold nanowire batteries to graphene-based batteries have shown great promise for the future – be it in terms of charging time or capacity. The increasing efficiency of computing devices and the improvement in battery technology means that everything is well balanced, but a revolutionary battery breakthrough is what the world needs right now.